Developer Pack

BrightCart



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Author	BrightCart team

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01

Developer Guide

Developer Pack
BrightCart

Developer Guide

BrightCart is a React 18 single-page storefront built with Vite and styled with Tailwind CSS (`package.json`). This guide takes you from a fresh clone to a running dev server.

Note

Dependencies were not installed in the environment that generated this guide, so the npm commands below were read from `package.json` but not executed. They are marked `untested`.

Prerequisites

Tool	Version	Source
Node.js	20	`.nvmrc` , `node-version: 20` in `.github/workflows/ci.yml`
npm	bundled with Node 20	CI runs `npm ci`
Git	any recent	—

Clone to running

1. Clone the repository and enter it. The remote URL is not recorded in the repo `assumption`.

```
git clone <repository-url> brightcart
cd brightcart
```

2. Create your environment file and fill in the values:

```
cp .env.example .env
```

Variable	Needed for
`VITE_API_URL`	Backend base URL used by `src/pages/Home.tsx`
`VITE_SENTRY_DSN`	Error reporting (`src/main.tsx`); empty disables it
`VITE_STRIPE_PUBLISHABLE_KEY`	Checkout (`src/pages/Checkout.tsx`)

3. Install dependencies untested:

```
npm install
```

4. Start the dev server untested:

```
npm run dev
```

Vite serves the app on `http://localhost:5173` by default assumption. No port is set in the repo.

Project structure

```
brightcart/
├── index.html            # Vite entry; loads Poppins + Playfair Display
├── tailwind.config.js    # brand colours and font families
├── vercel.json           # SPA rewrite: every path serves /
├── public/               # logo.svg, favicon.svg
├── src/
│   ├── main.tsx          # React root + Sentry.init
│   ├── App.tsx           # routes: / and /checkout
│   ├── index.css         # Tailwind layers + CSS variables
│   ├── pages/            # Home.tsx, Checkout.tsx
│   ├── components/       # ProductCard.tsx
│   └── App.test.tsx      # Vitest + Testing Library
└── .github/workflows/ci.yml
```

Common workflows

Command	What it does	Tested
`npm run dev`	Start the Vite dev server	untested
`npm run build`	Type-check (`tsc -b`) and build to `dist/`	untested
`npm run preview`	Serve the production build locally	untested
`npm run lint`	ESLint over the repo	untested
`npm test`	Run Vitest once (`vitest run`)	untested

Add a page

1. Create `src/pages/<Name>.tsx` , following `src/pages/Home.tsx` .
2. Register a route in `src/App.tsx` :

```
<Route path="/orders" element={<Orders />} />
```

3. Production needs no server change: `vercel.json` already rewrites every path to `/`.

Use the brand styles

Use the Tailwind tokens from `tailwind.config.js` rather than raw hex values: `bg-brand`, `text-brand-700`, `font-display` (Playfair Display) and `font-sans` (Poppins).

Testing

```
npm test
```

`src/App.test.tsx` has 2 tests: the home page renders, and a placeholder for the checkout route. See Testing Strategy.

Debugging tips

- Vite only exposes variables prefixed with `VITE_` to the browser. A missing value reads as `undefined` rather than throwing, so check `.env` first when a feature silently fails.
- Sentry starts in `src/main.tsx` with `VITE_SENTRY_DSN`. Leave it empty locally to avoid sending development errors.
- `loadStripe` in `src/pages/Checkout.tsx` runs at module load. An invalid publishable key shows up in the browser console as soon as the checkout page is imported.

02

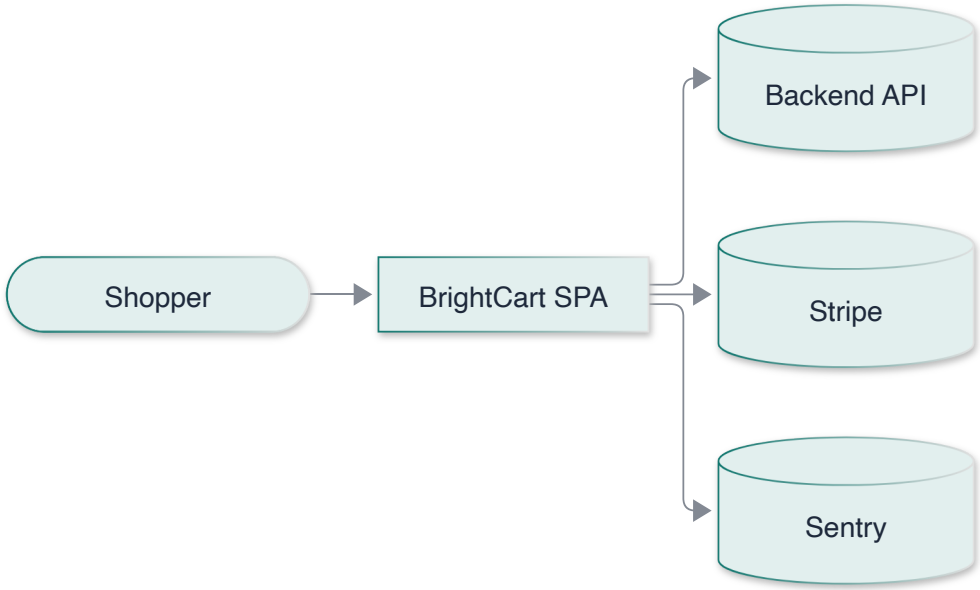
Architecture

Architecture

Overview

BrightCart is a client-side React single-page application. Vite builds it into static files that Vercel serves. The browser talks directly to three external systems: BrightCart's own backend API (`VITE_API_URL`), Stripe for payments and Sentry for error reporting. There is no server code in this repository.

System context



External system	Used for	Evidence
Backend API	Product data (planned)	`VITE_API_URL` read in `src/pages/Home.tsx`
Stripe	Payments	`loadStripe` in `src/pages/Checkout.tsx`
Sentry	Error reporting	`Sentry.init` in `src/main.tsx`
Vercel	Hosting and deploys	`vercel.json` , `deploy` job in `.github/workflows/ci.yml`

Components

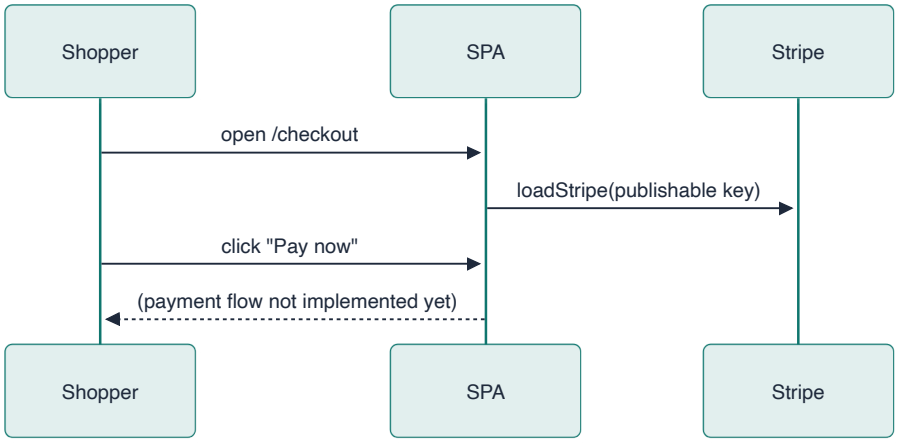
Component	Responsibility	Tech	Location
App shell	Router and page layout	React Router 6	`src/App.tsx`

02

Architecture

Component	Responsibility	Tech	Location
Home page	Product listing	React, Tailwind	`src/pages/Home.tsx`
Checkout page	Starts payment	Stripe.js	`src/pages/Checkout.tsx`
ProductCard	Product tile	React	`src/components/ProductCard.tsx`
Bootstrap	Mounts React, starts Sentry	React DOM	`src/main.tsx`

Data flow: checkout



1. React Router renders `Checkout` for `/checkout` (`src/App.tsx`).
2. Importing the page calls `loadStripe` with `VITE_STRIPE_PUBLISHABLE_KEY` .
3. The **Pay now** button has no handler yet, so no payment is created assumption that this is work in progress.

Deployment view

GitHub Actions runs `lint` and `test` on every push and pull request. On `main` it also runs `npx vercel deploy --prod` with the `VERCEL_TOKEN` secret. `vercel.json` rewrites every path to `/` , so client-side routes survive a page refresh.

Cross-cutting concerns

Concern	Approach	Evidence
Configuration	Build-time `VITE_*` env vars	`.env.example`
Error handling	Sentry browser SDK	`src/main.tsx`

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Architecture

Concern	Approach	Evidence
Styling	Tailwind with brand tokens	`tailwind.config.js`
Authentication	None in the frontend	no auth code found

Decision log

ADR-001: Single-page app on Vite

- **Status:** Accepted
- **Context:** The storefront needs fast navigation and no server rendering assumption.
- **Decision:** React 18 + Vite, routed client-side (`package.json` , `src/App.tsx`).
- **Consequences:** Static hosting is enough. SEO for product pages is limited without pre-rendering.

ADR-002: Tailwind CSS for styling

- **Status:** Accepted
- **Context:** One consistent brand palette across components.
- **Decision:** Tailwind 3 with brand colours and fonts in `tailwind.config.js` .
- **Consequences:** No separate CSS files per component; styles live in class names.

ADR-003: Vercel hosting

- **Status:** Accepted
- **Context:** Static SPA with preview needs assumption.
- **Decision:** Deploy with the Vercel CLI from CI (`.github/workflows/ci.yml`).
- **Consequences:** Deploys depend on the `VERCEL_TOKEN` secret and Vercel availability.

03

Assumptions and open questions

Assumptions and open questions

Not included: API Reference, Data Model, Contributing Guide, Testing Strategy, CI/CD and Deployment, Environment and Configuration Reference, Security Threat Model, Dependency Inventory, Tech Debt Register, Integration Guide (not selected for this sample).

#	Chapter	Item	Type	Why unverified	Who can confirm
A1	Developer Guide	Repository URL	assumption	no git remote in the repo	maintainer
A2	Developer Guide	Dev server port 5173	assumption	Vite default, not configured	frontend lead
A3	Developer Guide	npm commands work as listed	assumption	dependencies not installed during generation	frontend lead
Q1	Developer Guide	Where does the backend behind `VITE_API_URL` live?	open question	not in this repo	engineering lead
A4	Architecture	Pay now is unfinished work, not a bug	assumption	button has no handler	frontend lead
A5	Architecture	Reasons behind ADR-001 and ADR-003	assumption	inferred from code, no ADR files	tech lead
Q2	Architecture	What API does `VITE_API_URL` point to, and who owns it?	open question	backend not in repo	engineering lead